6.1.1. Incremented Date

Algorithm:

Step 1: Start

Step 2: Input day

Step 3: Input month

Step 4: Input year

Step 5: Set flag = True

Step 6: If (year < 1) OR (month < 1) OR (month > 12)
    Set flag = False

Step 7: If month is 1, 3, 5, 7, 8, 10, or 12
    Set maxday = 31

Step 8: Else if month is 4, 6, 9, or 11
    Set maxday = 30

Step 9: Else (month is February)
    If (year % 4 == 0)
        Set maxday = 29
    Else
        Set maxday = 28

Step 10: If (day < 1) OR (day > maxday)
    Set flag = False

Step 11: Increase day by 1
    day = day + 1

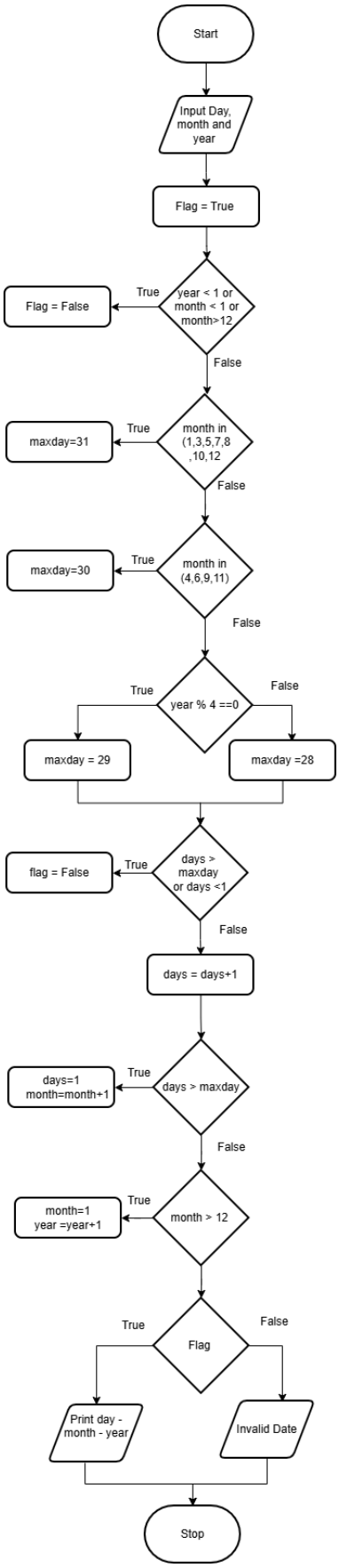
Step 12: If day > maxday
    Set day = 1
    Set month = month + 1

Step 13: If month > 12
    Set month = 1
    Set year = year + 1

Step 14: If flag == True
    Print the date in format DD-MM-YYYY
  Else
    Print "Invalid Date"

Step 15: Stop

Flowchart:


Start
Input Day, month and year
Flag = True
year < 1 or month < 1 or month>12
True
Flag = False
False
month in (1,3,5,7,8 ,10,12
True
maxday=31
False
month in (4,6,9,11)
True
maxday=30
False
year % 4 ==0
True
False
maxday = 29
maxday =28
days > maxday or days <1
True
flag = False
False
days = days+1
days > maxday
True
days=1 month=month+1
False
month > 12
True
month=1 year =year+1
Flag
True
False
Print day - month - year
Invalid Date
Stop

CODETANTRA Home aryan.marghade.batch2025@sitnagpur.siu.edu.in Support Logout

6.1.1. Incremented Date

29:13

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. **Valid Month**: 1 to 12
2. **Valid Day**: Depends on the month and year
   - January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
   - April (4), June (6), September (9), November (11): 1 to 30 days
   - February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
3. **Valid Year**: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Output Format:

- If the date is valid, print the incremented date in the format:

```
<DD>-<MM>-<YYYY>
```

- If the date is invalid, print: "Invalid Date".

Sample Test Cases +

Explorer

nextDate.py Submit

```python
year=int(input())
flag = True
if (year < 1 or month < 1 or month>12 ):
    flag = False

if( month in (1,3,5,7,8,10,12)):
    maxday=31
elif(month in (4,6,9,11)):
    maxday=30
else:
    if(year % 4 ==0):
        maxday = 29
    else:
        maxday =28

if (days > maxday or days <1):
    flag = False
```

Debugger

Average time	Maximum time
0.015 s	0.033 s
15.50 ms	33.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 23 ms Debug

Expected output	Actual output
15	15
3	3
2024	2024
16-03-2024	16-03-2024

Test case 2 9 ms

Terminal Test cases

Prev Reset Submit Next